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ON

Federated Learning for Privacy-Preserving Collaborative AI: An Experimental Analysis of Data Distribution

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In

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Federated Learning for Privacy-Preserving Collaborative AI: An Experimental Analysis of Data Distribution"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gurpreet Singh**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

This project focuses on the implementation and analysis of Federated Learning (FL) as a privacy-preserving machine learning approach. Traditional machine learning systems rely on centralized data collection, which raises concerns related to data privacy, security, and regulatory compliance. Federated Learning addresses these issues by enabling decentralized training, where data remains on local devices and only model updates are shared.

In this project, a configurable federated learning simulation system was developed using Python and TensorFlow. The system supports both Independent and Identically Distributed (IID) and Non-IID data scenarios. Multiple clients were simulated, and model training was performed locally on each client. The global model was updated using the Federated Averaging (FedAvg) algorithm.

Experiments were conducted to compare centralized learning, IID federated learning, and Non-IID federated learning. The results showed that federated learning performs close to centralized learning under IID conditions, achieving high accuracy. However, in Non-IID scenarios, performance drops due to data heterogeneity and client drift.

The project also includes visualization of accuracy trends, convergence behavior, and data distribution across clients. These insights help in understanding the practical challenges of federated learning systems.

Overall, this project demonstrates the trade-offs between privacy and performance in decentralized machine learning and provides a hands-on understanding of federated learning concepts.

INTRODUCTION

With the rapid growth of Artificial Intelligence (AI) and data-driven systems, large volumes of data are being generated from various sources such as smartphones, IoT devices, sensors, and web applications. Traditionally, machine learning models are trained using centralized approaches, where all data is collected and stored on a central server. While this approach is effective in achieving high model accuracy, it introduces serious concerns related to data privacy, security risks, and regulatory compliance.

In many real-world domains such as healthcare, finance, and personalized applications, user data is highly sensitive and cannot be shared freely. Centralized storage of such data increases the risk of data breaches and creates a single point of failure. This limitation has led to the development of Federated Learning (FL), which provides a decentralized alternative to traditional machine learning.

Federated Learning allows models to be trained locally on user devices without transferring raw data to a central server. Instead, only model updates such as weights or gradients are shared. These updates are then aggregated to form a global model. This approach helps in preserving data privacy while still enabling collaborative learning.

However, federated learning introduces new challenges. One of the major challenges is data heterogeneity, commonly referred to as Non-IID data distribution. In real-world scenarios, data across different users is not uniform, which makes model training more complex. Another challenge is communication overhead, as multiple rounds of interaction are required between clients and the central server.

In this project, we aim to build a federated learning simulation system and analyze its behavior under different data distributions. We compare centralized learning with federated learning in both IID and Non-IID settings. The objective is to understand how data

distribution affects model performance, convergence behavior, and training stability.

This project not only demonstrates the implementation of federated learning but also provides a detailed analysis of its practical limitations and advantages.

METHODOLOGY

The methodology of this project is focused on designing and analyzing a federated learning system in a controlled experimental setup. The goal was to compare different learning approaches and observe how data distribution affects performance.

The overall workflow of the project can be divided into the following stages:

- Centralized Learning
- Federated Learning (IID Setup)
- Federated Learning (Non-IID Setup)

In the centralized learning phase, the model was trained using the complete dataset. This acts as a baseline since the model has access to all data and can learn optimal patterns.

In the federated learning setup, multiple clients were simulated. Each client trained the model locally using its own data. The training was performed in rounds, where each round consisted of:

- Sending the global model to all clients
- Performing local training on each client
- Collecting updated model weights
- Aggregating weights using the FedAvg algorithm

Two types of data distributions were considered:

- **IID (Independent and Identically Distributed):**
  - Data is randomly and evenly distributed
  - Each client has similar data patterns
  - Represents an ideal scenario

- **Non-IID (Non-Identically Distributed):**
  - Data is unevenly distributed across clients
  - Each client contains biased data
  - Represents real-world conditions

The Federated Averaging (FedAvg) algorithm was used for aggregation. It computes the average of model weights received from all clients and updates the global model accordingly.

To evaluate the system, the following metrics were used:

- Model Accuracy
- Training Convergence
- Number of Communication Rounds
- Training Time

Additionally, visualization techniques were used to analyze performance differences between IID and Non-IID setups.

This methodology helps in understanding how federated learning behaves in both ideal and realistic conditions.

TOOLS AND TECHNOLOGIES

The project was developed using a combination of programming tools and machine learning frameworks to simulate and analyze federated learning systems.

The main tools and technologies used are:

- **Python**
  - Primary programming language
  - Easy to use and widely supported in machine learning

- TensorFlow and Keras
  - Used for building and training neural networks
  - Provides high-level APIs for model design
  - Efficient for deep learning tasks

- **NumPy**
  - Used for numerical computations
  - Helps in handling arrays and dataset manipulation
  - Used during weight aggregation in federated learning

- **Matplotlib**
  - Used for plotting graphs
  - Helps visualize accuracy, loss, and convergence

- **Seaborn**
  - Used for advanced data visualization
  - Helps in visualizing data distribution across clients

- **Jupyter Notebook / Google Colab**
  - Used as the development environment
  - Allows interactive coding and visualization
  - Makes experimentation easier

- **MNIST Dataset**
  - Standard dataset of handwritten digits
  - Used for training and evaluation
  - Simple and effective for benchmarking

These tools collectively helped in building a flexible and modular federated learning system.

IMPLEMENTATION

The implementation of this project focuses on building a modular federated learning simulation system. The system was designed to be flexible and configurable, allowing different experiments to be performed easily.

The implementation process can be divided into the following steps:

1. Data Preparation

The MNIST dataset was loaded and preprocessed. The preprocessing steps included:

- Normalizing pixel values to improve training stability
- Reshaping images to match CNN input format
- Splitting data into training and testing sets

2. Model Design

A Convolutional Neural Network (CNN) was implemented using TensorFlow. The model architecture includes:

- Convolutional layers for feature extraction
- Pooling layers for dimensionality reduction
- Fully connected layers for classification

The model was kept simple to focus on training behavior rather than complexity.

3. Centralized Training

In the centralized setup:

- The model was trained using the full dataset
- Training was performed for multiple epochs
- Accuracy and loss were recorded

This setup provides a baseline for comparison.

4. Client Simulation

Multiple clients were simulated in the system. Each client was assigned a portion of the dataset.

Two types of client configurations were implemented:

• **IID Clients:**
- Data was randomly distributed
- Each client had similar data distribution

• **Non-IID Clients:**
- Data was split based on labels
- Each client had specific classes only

5. Federated Training Process

The federated training followed a round-based approach:

• The global model was sent to all clients
• Each client trained the model locally for one epoch
• Local model weights were sent back to the server
• The server aggregated weights using FedAvg

This process was repeated for multiple rounds.

6. Weight Aggregation (FedAvg)

The aggregation step involved:

• Collecting weights from all clients
• Computing the average of corresponding weights
• Updating the global model

7. Performance Tracking

During training, the following were recorded:

• Accuracy after each round
• Loss values
• Training time

8. Visualization

Several graphs were generated to analyze the system:

• Accuracy vs Rounds

- Loss vs Rounds
- Accuracy comparison (Centralized vs IID vs Non-IID)
- Data distribution across clients
- Convergence behavior

These visualizations helped in understanding how different setups affect performance.

Overall, the implementation provides a complete pipeline for simulating and analyzing federated learning systems in a structured and practical way.

MAJOR FINDINGS / OUTCOMES / RESULTS

The experiments conducted in this project provided clear insights into the behavior of federated learning under different data distributions. The system was evaluated using three setups: centralized learning, federated learning with IID data, and federated learning with Non-IID data.

1. Centralized Learning Results

The centralized model achieved the highest accuracy among all setups.

- Final Accuracy: ~99.15%
- Training Time: ~167 seconds
- Convergence: Fast and stable

Observations:

- The model had access to the complete dataset
- Learning was consistent across all classes
- Convergence was smooth with minimal fluctuations

This setup serves as the performance benchmark for comparison.

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2. Federated Learning (IID) Results

In the IID setup, data was evenly distributed across clients.

- Final Accuracy: ~99.19%
- Training Time: ~174 seconds
- Convergence: Slightly slower than centralized

Observations:

- Accuracy was very close to centralized learning
- The model required multiple communication rounds
- Training remained stable across rounds

Round-wise Accuracy Trend:

- Round 1: ~97.6%
- Round 3: ~98.9%
- Round 5: ~99.19%

Key Insight:

- Federated learning performs almost equally well when data is evenly distributed

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3. Federated Learning (Non-IID) Results

In the Non-IID setup, each client had biased data.

- Final Accuracy: ~94.03%
- Training Time: ~199 seconds
- Convergence: Slower and less stable

Observations:

- Accuracy dropped compared to IID setup
- Training showed fluctuations in early rounds
- Convergence required more rounds

Round-wise Accuracy Trend:

- Round 1: ~73.3%
- Round 2: ~86.0%

- Round 3: ~90.4%
- Round 5: ~94.0%

Key Insight:

- Data imbalance leads to inconsistent model updates
- This affects global model performance

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4. Comparison Summary

The comparison between all three approaches shows clear differences:

- Centralized learning provides the highest accuracy
- IID federated learning performs very close to centralized
- Non-IID federated learning shows reduced accuracy

Main Differences:

- Data Availability:
  - Centralized: Full data access
  - Federated: Distributed data

- Performance:
  - IID: High performance
  - Non-IID: Performance drop

- Training Stability:
  - IID: Stable
  - Non-IID: Fluctuating

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5. Convergence Behaviour

The convergence patterns observed:

- **IID setup:**
  - Smooth improvement across rounds
  - Stable learning

• **Non-IID setup:**
- Slower improvement
- Fluctuations in accuracy

This confirms that data distribution plays a major role in training dynamics.

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6. Communication Overhead

Federated learning introduces additional cost due to communication rounds.

Observations:

• Multiple rounds required for convergence
• Increased training time compared to centralized
• Overhead grows with number of clients

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7. Visualization Insights

From the graphs generated:

• **Accuracy graph shows:**
- IID curve close to centralized
- Non-IID curve lower and less stable

• **Loss graph shows:**
- Non-IID has higher loss values
- IID stabilizes faster

• **Data distribution graph shows:**
- Clear imbalance in Non-IID setup

Overall, the results clearly demonstrate that while federated learning improves privacy, its performance is highly dependent on data distribution and system design.

CONCLUSION AND FUTURE SCOPE

Conclusion

In this project, we implemented a federated learning simulation system and compared its performance with centralized learning. The study focused on understanding how data distribution affects model performance and convergence.

The results show that federated learning performs very well in IID conditions, achieving accuracy close to centralized learning. However, in Non-IID scenarios, the performance drops due to data heterogeneity and inconsistent updates from different clients.

Some key conclusions from the project are:

- Federated learning is effective for privacy-preserving machine learning
- IID data distribution leads to stable and high-performance models
- Non-IID data introduces challenges such as:
  - Client drift
  - Slower convergence
  - Reduced accuracy

- Communication overhead is an important factor in federated systems

Overall, the project provides a clear understanding of both the advantages and limitations of federated learning.

Future Scope

This project can be extended further in multiple ways to improve its functionality and realism.

Some possible future improvements include:

- Implementing advanced algorithms:
  - FedProx
  - SCAFFOLD
  - Personalized Federated Learning

- Increasing the number of clients to simulate large-scale systems

- Using more complex datasets such as:
  - CIFAR-10
  - Real-world datasets

- Adding real-time visualization dashboards

- Implementing secure aggregation techniques for enhanced privacy

- Deploying the system in a distributed environment instead of simulation

- Integrating the system with real applications such as:
  - Healthcare data systems
  - Mobile applications

These improvements can make the system more realistic and suitable for real-world deployment.

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APPENDICES

Appendix A: Source Code

The complete implementation of the federated learning system is provided in the Jupyter Notebook submitted along with this report.

Appendix B: Experimental Outputs

This includes:

- Accuracy logs
- Training outputs
- Graphs and visualizations

Appendix C: System Configuration

- Number of Clients: 3
- Number of Rounds: 5
- Dataset Used: MNIST

- Model Type: CNN